Test Method for Durometer Hardness	Test method # VF-A01 REF # : TM-R- 012 Page 1 of 4
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PURPOSE AND SCOPE: To measure the hardness of durometer of a test material. This is applicable for foam, plastic and rubber.

REFERENCE: ASTM D2240

EQUIPMENT AND MATERIAL:

1. Durometer hardness testers and sample thickness according to the following guidelines:

   Durometer Hardness A - Solid rubber material (6 mm thick, minimum)

   Durometer Hardness D - Solid plastic material (6 mm thick, minimum)

   Durometer Hardness C - Medium to hard foam material (10 mm thick, minimum)

   Durometer Hardness F - Soft foam material (10 mm thick, minimum)

2. Constant load stand B
3. Steel scale, capable of measuring to 0.5 mm.
4. Foam, plastic , rubber

SAMPLE PREPARATION:

Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A) General (all materials except molded EVA midsoles and unit-soles).

The sample must have the minimum thickness as stated above and be large enough to perform at least six measurements from different areas on the test sample, with no measurement being done within 12 mm of an edge. The sample must be flat, so that the presser foot of the durometer hardness tester can lie flat on the surface of the test material

Note: PU midsole and Die-Cut EVA midsole sampling and measurement must be performed according to the requirements of Appendix C.

B) Compression-molded and Injection-molded EVA midsole and unit sole:

Using a midsole or unit-sole, the samples are prepared by cutting the midsole or unit-soles from heel to toe through the thickness portion of both the heel and forefoot. The cut is made perpendicular to the top and bottom surfaces, and such that a smooth, flat surface is produced for measuring durometer. The cut must be at least 10 mm from the edge of the sample. Heel

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thickness must be a minimum of 12 mm and forefoot thickness must be a minimum of 8 mm. With the use of a steel scale, determine the total thickness of both the heel and forefoot by measuring from the top skin to the bottom skin. Divide the total thickness by 2 to determine the centerline location.

Next, lines should be drawn along the centerline of the thickest section in both the heel and forefoot .The line should be 20 -40 mm long.

Test plaques (preferably for development material only). The cut is made perpendicular to the top and bottom surfaces, and such that a smooth, flat surface is produced for measuring durometer, the cut must be at least 20 mm from any edge of the test plaque, test plaques must be a minimum of 10 mm in thickness.

With the use of a steel scale, determine the total thickness of the sample by measuring from the top skin to the bottom skin. Divide the total thickness by 2 to determine the centerline location, next, line should be drawn along the centerline of the test plaque over the entire length of the cut.

Additionally, lines should be drawn 3-4 mm below both the top and bottom surfaces of the test plaques.

REPETITIONS: Perform six measurements per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory environment @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A) General (all materials except molded EVA midsole and unit-soles).

1. Confirm proper operation of the durometer using the test block.
2. Place the test sample on a hard horizontal surface. Sample must be at preconditioned at the temperature of 23.0 ± 2.0˚C.
3. Apply the presser foot of the durometer hardness tester to the test sample as rapidly as possible (without shock), keeping the foot parallel to the surface of the test sample.
   The following guidelines should be used to achieve consistent pressure loadings:

Durometer	Loading
Durometer Hardness D	5kg
Durometer Hardness A	1kg
Durometer Hardness C	1kg
Durometer Hardness F	No additional load

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4. Take the reading within **three** seconds, or if the durometer has a maximum indicator, use the maximum reading. Record the reading to nearest unit on the durometer gauge.
5. Reading must be taken at least 12 mm from any edge, and at least 6 mm apart.

REPORT: The range of durometer readings to the nearest unit of the specific durometer being used.

B) Compression-molded and injection-molded midsoles and unit soles (see section C for test plaques).

1. Confirm proper operation of the durometer using the test block.
2. Place the test sample on a hard horizontal surface, sample must be preconditioned at temperature of 23.0±2.0°C.

Note: *If sample is not flat and needs to be hand held to take the measurement, it should be done by ensuring that there is at least 10mm of foam below the durometer presser foot, and such that the surface being measured does not bend.*

3. Apply the presser foot of the durometer to the test sample as rapidly as possible (without shock), keeping the foot parallel to the surface of the test sample.
4. To determine the durometer of the midsole or unit-sole, take 3-6 measurements along the centerline of the thickest portion of the heel.
5. To determine proper cure of the midsole or unit-sole, take 3-6 measurements along the centerline of the thickness portion of the forefoot.

REPORT:

The durometer of the midsole or unit-sole, as the range of measurements which were taken along the centerline of the thickest portion of the forefoot.

Note: *To determine that the midsole or unit-sole has been properly cured, determine the average of the centerline reading that were taken in both the heel and forefoot. Subtract the average heel centerline durometer from the average forefoot centerline durometer. (Requirements are indicated on each VFA Specification)*

C) Compression-molded and injection-molded midsole and unit-sole test plaques.

1. Confirm proper operation of the durometer using the test block.
2. Place the test sample on a hard horizontal surface. Sample must be preconditioned at temperature of 23.0±2.° C. **Note:** If sample is not flat and needs to be hand held to take the measurement, it should be done by ensuring that there is at least 10mm of foam below the durometer presser foot, and such that the surface being measured does not bend.
3. Apply the presser foot of the durometer hardness tester to the test sample as rapidly as possible (without shock), keeping the foot parallel to the surface of the test sample.

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4. To determine the durometer of the midsole or unit-sole test plaque, take 6 measurements along the centerline of the test plaque (reading must be at least 2 cm apart and taken across the entire length of the sample).
5. To determine proper cure of the midsole or unit-sole test plaque take 6 measurements along the line 3-4mm below the top surface of the test plaque, and then 6 measurements along the line 3-4 above surface of the test plague . Readings must be at least 2 cm apart and taken along the entire length of the sample.

(Note. *The presser foot of the durometer hardness tester must not contact the skin or a false reading will result).*

REPORT:

The durometer of the midsole or unit-sole test plaque, as the range of measurement which were taken along the centerline.

Note: *To determine that the midsole or unit-sole test plaque has been properly cured, determine the average of the centerline durometer reading and the average of the reading taken 3-4 mm from both the top and bottom surface of the test plaque. Subtract the average centerline durometer from the average durometer measured 3-4 mm from the top surface. Then, subtract the average centerline durometer from the average durometer measured 3-4 mm from the bottom surface. (Requirements are indicated on each VFA Footwear Lab Specification.)*